

**EXPLORATORY DATA ANALYSIS**

**USING POWER BI**

**REPORT**

**WEEK – 3**

**FINANCIAL DATA IMPORT, TRANSFORMATION**

**&**

**TIME INTELLIGENCE**

**SUBMITTED BY:**

**Divya Dharshni P**

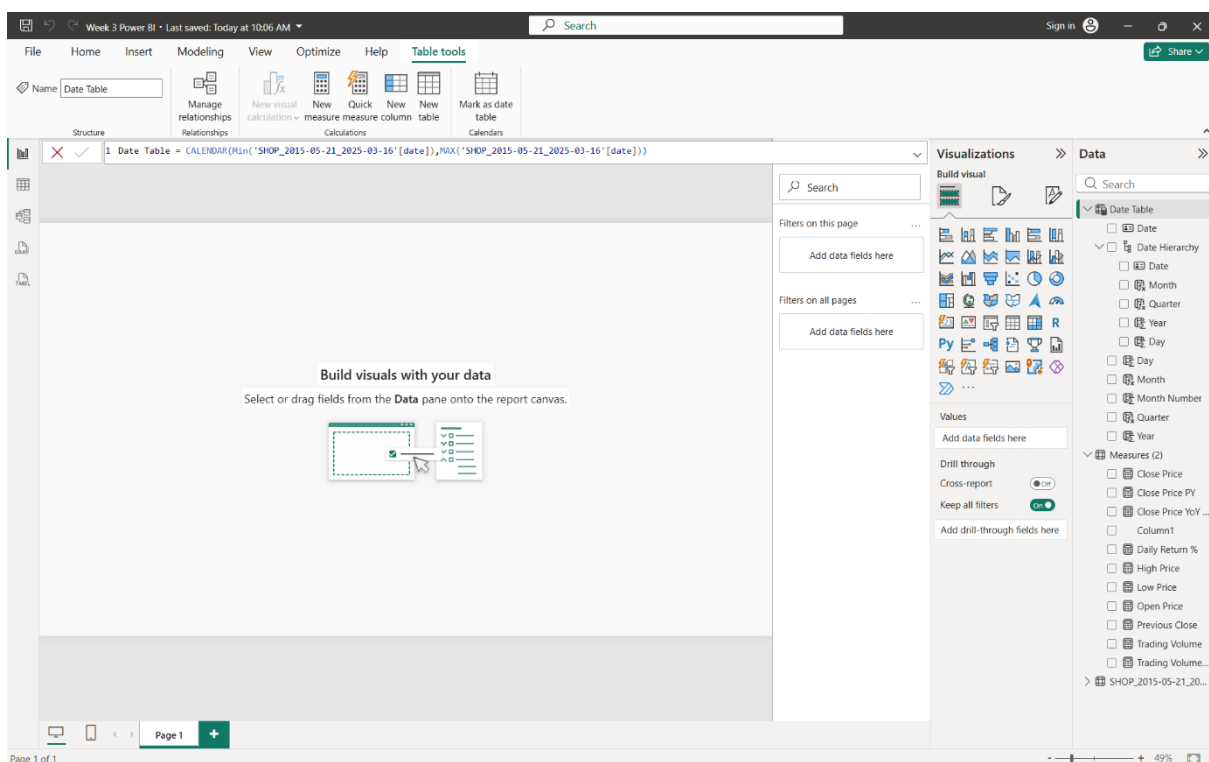
**E25DA014**

## 1. Introduction

The objective of this task was to analyze Shopify Stock Data (NYSE: SHOP) from 2015-2025. The dataset contains daily stock metrics like Open, High, Low, Close and Volume. The goal was to build a time-intelligence enabled data model in Power BI.

## 2. Data Preparation (Power Query)

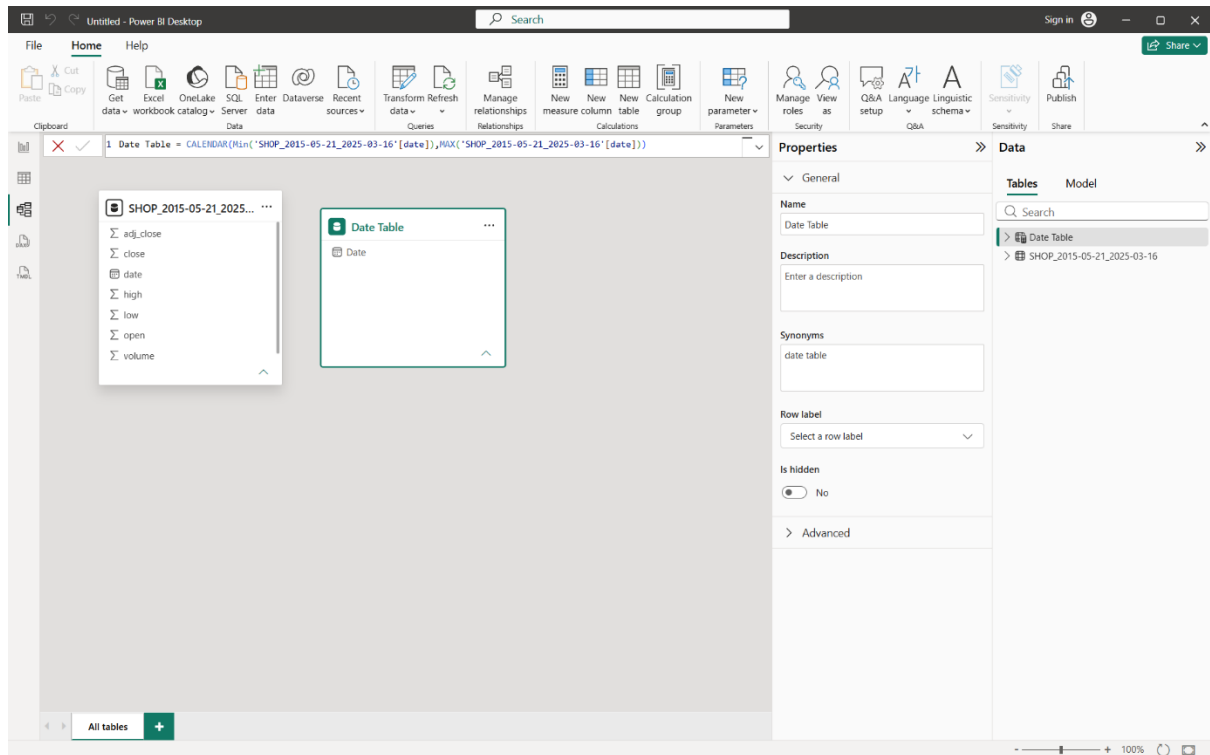
- Imported Shopify stock CSV file into Power Query.
- Changed Data Type of Date column to Date format and removed time component.
- Sorted data by Date in Ascending order.
- Ensured numeric columns (Open, High, Low, Close, Volume) were set to Decimal/Whole number.



## 3. Date Dimension Table :

A dedicated date table was created using DAX to enable time intelligence:

```
Date Table = CALENDAR(Min('SHOP_2015-05-21_2025-03-16'[date]),MAX('SHOP_2015-05-21_2025-03-16'[date]))
```



Additional columns added: Year, Month Number, Month, Quarter, Day.

1. `Day = DAY('Date Table'[Date])`
2. `Month = FORMAT('Date Table'[Date], "MMM")`
3. `Month Number = MONTH('Date Table'[Date])`
4. `Quarter = "Q" & FORMAT('Date Table'[Date], "Q")`
5. `Year = Year('Date Table'[Date])`

Week 3 Power BI - Last saved: Today at 10:03 PM

Search

File Home Help Table tools

Name Date Table

Manage relationships New measure Quick New New table Mark as date table

Structure Calculations Calendars

| Date                  | Year | Quarter | Month | Month Number | Day |
|-----------------------|------|---------|-------|--------------|-----|
| 5/21/2015 12:00:00 AM | 2015 | Q2      | May   | 5            | 21  |
| 5/22/2015 12:00:00 AM | 2015 | Q2      | May   | 5            | 22  |
| 5/23/2015 12:00:00 AM | 2015 | Q2      | May   | 5            | 23  |
| 5/24/2015 12:00:00 AM | 2015 | Q2      | May   | 5            | 24  |
| 5/25/2015 12:00:00 AM | 2015 | Q2      | May   | 5            | 25  |
| 5/26/2015 12:00:00 AM | 2015 | Q2      | May   | 5            | 26  |
| 5/27/2015 12:00:00 AM | 2015 | Q2      | May   | 5            | 27  |
| 5/28/2015 12:00:00 AM | 2015 | Q2      | May   | 5            | 28  |
| 5/29/2015 12:00:00 AM | 2015 | Q2      | May   | 5            | 29  |
| 5/30/2015 12:00:00 AM | 2015 | Q2      | May   | 5            | 30  |
| 5/31/2015 12:00:00 AM | 2015 | Q2      | May   | 5            | 31  |
| 6/1/2015 12:00:00 AM  | 2015 | Q2      | Jun   | 6            | 1   |
| 6/2/2015 12:00:00 AM  | 2015 | Q2      | Jun   | 6            | 2   |
| 6/3/2015 12:00:00 AM  | 2015 | Q2      | Jun   | 6            | 3   |
| 6/4/2015 12:00:00 AM  | 2015 | Q2      | Jun   | 6            | 4   |
| 6/5/2015 12:00:00 AM  | 2015 | Q2      | Jun   | 6            | 5   |
| 6/6/2015 12:00:00 AM  | 2015 | Q2      | Jun   | 6            | 6   |
| 6/7/2015 12:00:00 AM  | 2015 | Q2      | Jun   | 6            | 7   |
| 6/8/2015 12:00:00 AM  | 2015 | Q2      | Jun   | 6            | 8   |
| 6/9/2015 12:00:00 AM  | 2015 | Q2      | Jun   | 6            | 9   |
| 6/10/2015 12:00:00 AM | 2015 | Q2      | Jun   | 6            | 10  |
| 6/11/2015 12:00:00 AM | 2015 | Q2      | Jun   | 6            | 11  |
| 6/12/2015 12:00:00 AM | 2015 | Q2      | Jun   | 6            | 12  |
| 6/13/2015 12:00:00 AM | 2015 | Q2      | Jun   | 6            | 13  |
| 6/14/2015 12:00:00 AM | 2015 | Q2      | Jun   | 6            | 14  |
| 6/15/2015 12:00:00 AM | 2015 | Q2      | Jun   | 6            | 15  |
| 6/16/2015 12:00:00 AM | 2015 | Q2      | Jun   | 6            | 16  |
| 6/17/2015 12:00:00 AM | 2015 | Q2      | Jun   | 6            | 17  |
| 6/18/2015 12:00:00 AM | 2015 | Q2      | Jun   | 6            | 18  |
| 6/19/2015 12:00:00 AM | 2015 | Q2      | Jun   | 6            | 19  |
| 6/20/2015 12:00:00 AM | 2015 | Q2      | Jun   | 6            | 20  |
| 6/21/2015 12:00:00 AM | 2015 | Q2      | Jun   | 6            | 21  |
| 6/22/2015 12:00:00 AM | 2015 | Q2      | Jun   | 6            | 22  |
| 6/23/2015 12:00:00 AM | 2015 | Q2      | Jun   | 6            | 23  |

Table: Date Table (3,586 rows)

Data

Search

Date Table Measures (2) SHOP\_2015-05-21\_2025-03-16

**The table was then Marked as Date Table using Date column. This is mandatory for time-intelligence functions to work:**

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Clipboard Data Queries Relationships Calculations Parameters Security Q&A Language Linguistic schema Q&A Sensitivity Publish

1 Date Table = CALENDAR(MIN('SHOP\_2015-05-21\_2025-03-16'[date]),MAX('SHOP\_2015-05-21\_2025-03-16'[date]))

SHOP\_2015-05-21\_2025-03-16

adj\_close  
close  
date  
high  
low  
open  
volume

Date Table

Date  
Day  
Month  
Month Number  
Quarter

Add related tables  
New measure  
New column  
Refresh  
Manage relationships  
Select measures  
Delete from model  
Hide in report view  
✓ Mark as date table  
Unhide all  
Collapse all  
Expand all

Trading Volume

Properties

General

Name  
Date Table

Description  
Enter a description

Synonyms  
date table

Row label  
Select a row label

Is hidden  
No

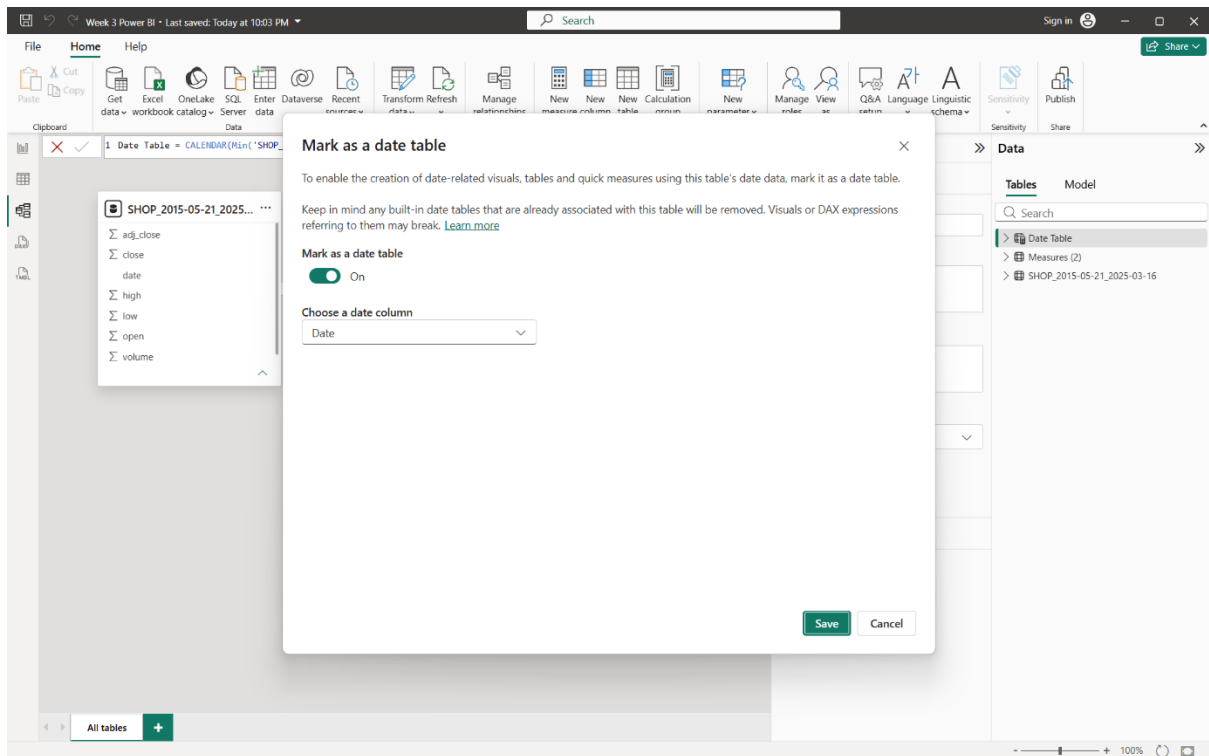
Advanced

Data

Search

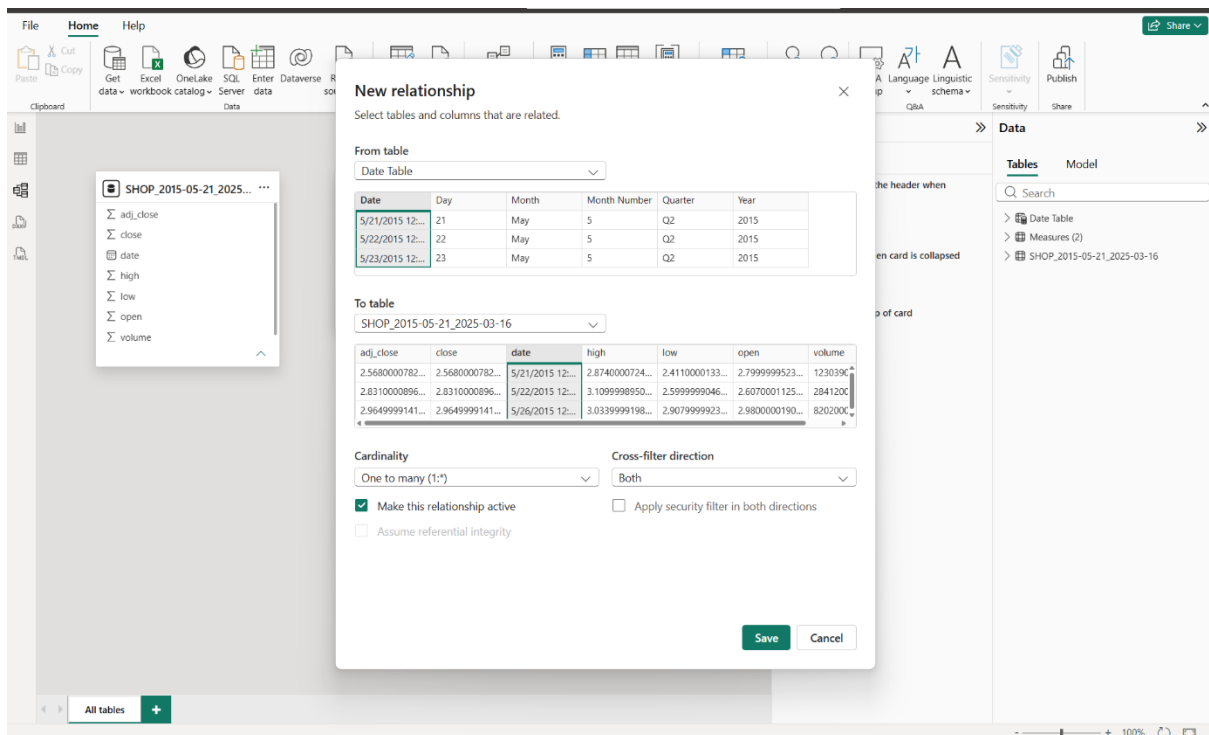
Date Table Measures (2) SHOP\_2015-05-21\_2025-03-16

All tables



## 4. Relationship Created:

Date Table[Date] (1) -> 'SHOP\_2015-05-21\_2025-03-16' [Date] (\*) - Active, both direction

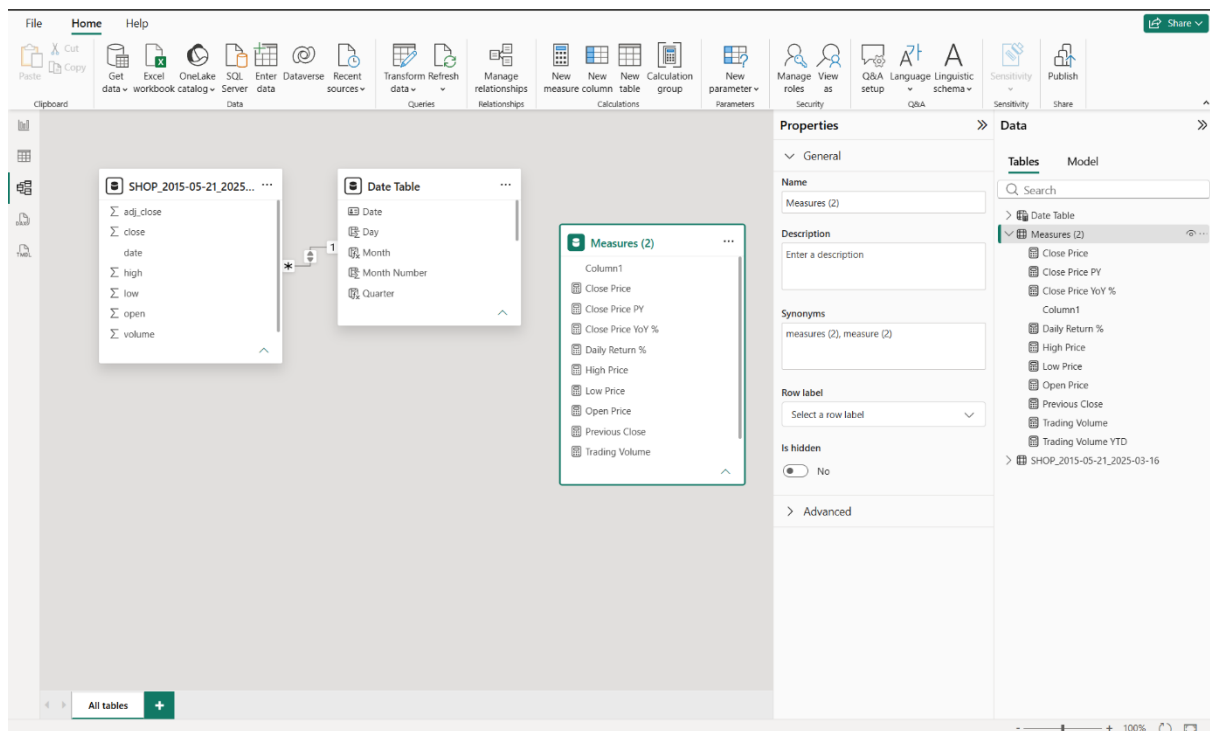


## 5. Core DAX Measures Created:

- ✓ Close Price = AVERAGE('SHOP\_2015-05-21\_2025-03-16' [Close])
- ✓ Open Price = AVERAGE('SHOP\_2015-05-21\_2025-03-16' [Open])
- ✓ High Price = MAX('SHOP\_2015-05-21\_2025-03-16' [High])
- ✓ Low Price = MIN('SHOP\_2015-05-21\_2025-03-16' [Low])
- ✓ Trading Volume = SUM('SHOP\_2015-05-21\_2025-03-16' [Volume])
- ✓ Previous Close = CALCULATE([Close Price],DATEADD('Date Table'[Date], -1, DAY))
- ✓ Daily Return % = DIVIDE([Close Price] - [Previous Close], [Previous Close])

## 6. Time Intelligence Measures:

- ✓ Trading Volume YTD = TOTALYTD([Trading Volume],'Date Table'[Date])
- ✓ Close Price PY = CALCULATE([Close Price],SAMEPERIODLASTYEAR('Date Table'[Date]))
- ✓ Close Price YoY % = DIVIDE([Close Price] - [Close Price PY],[Close Price PY],0)

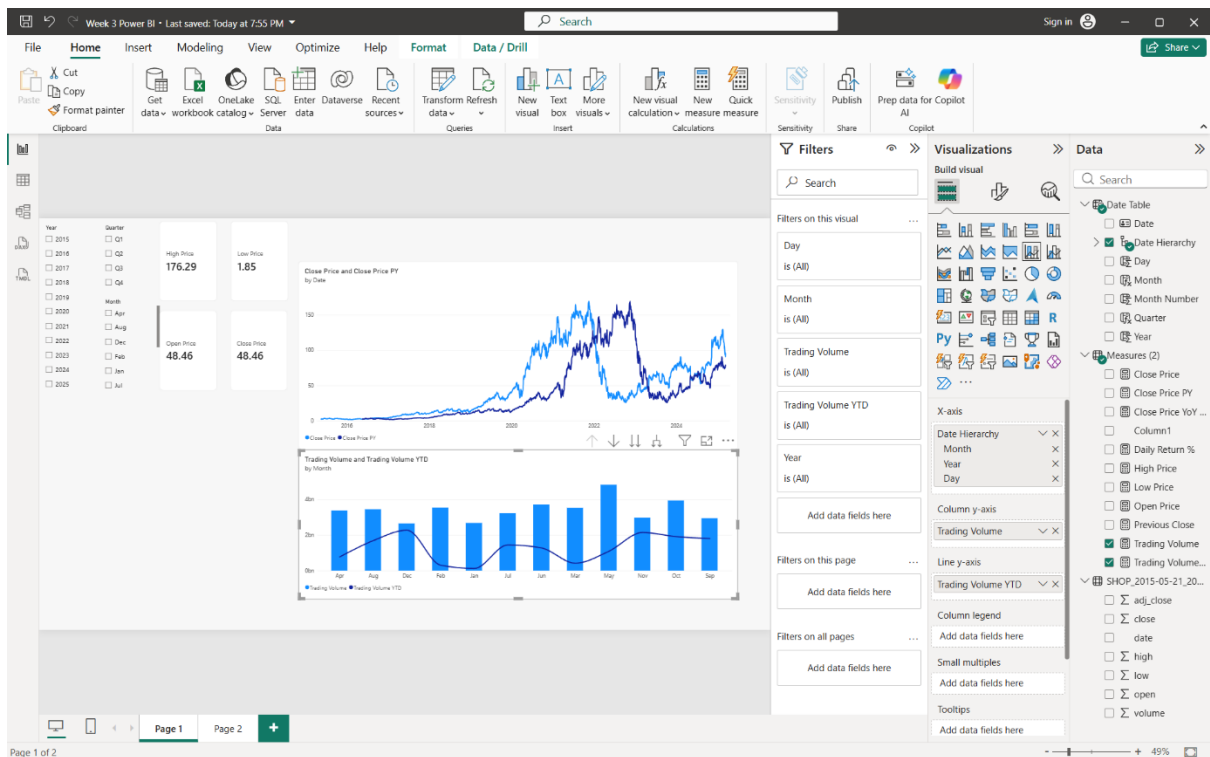
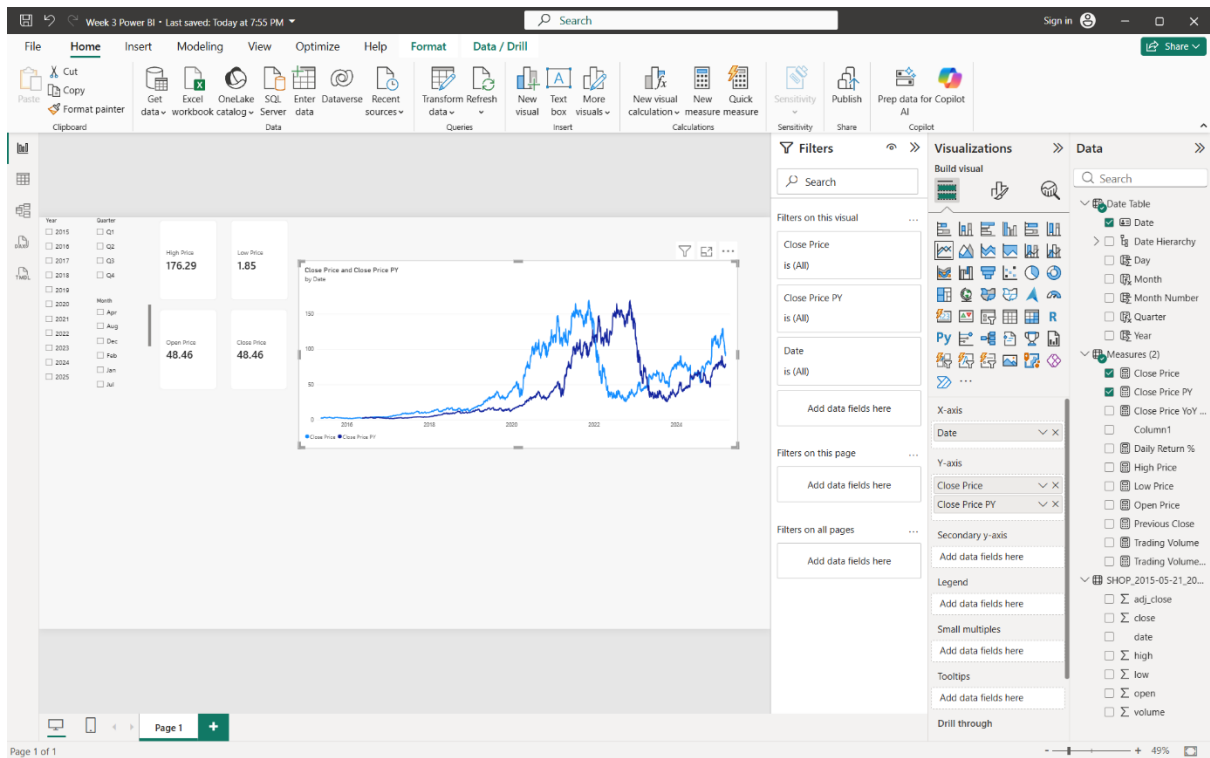


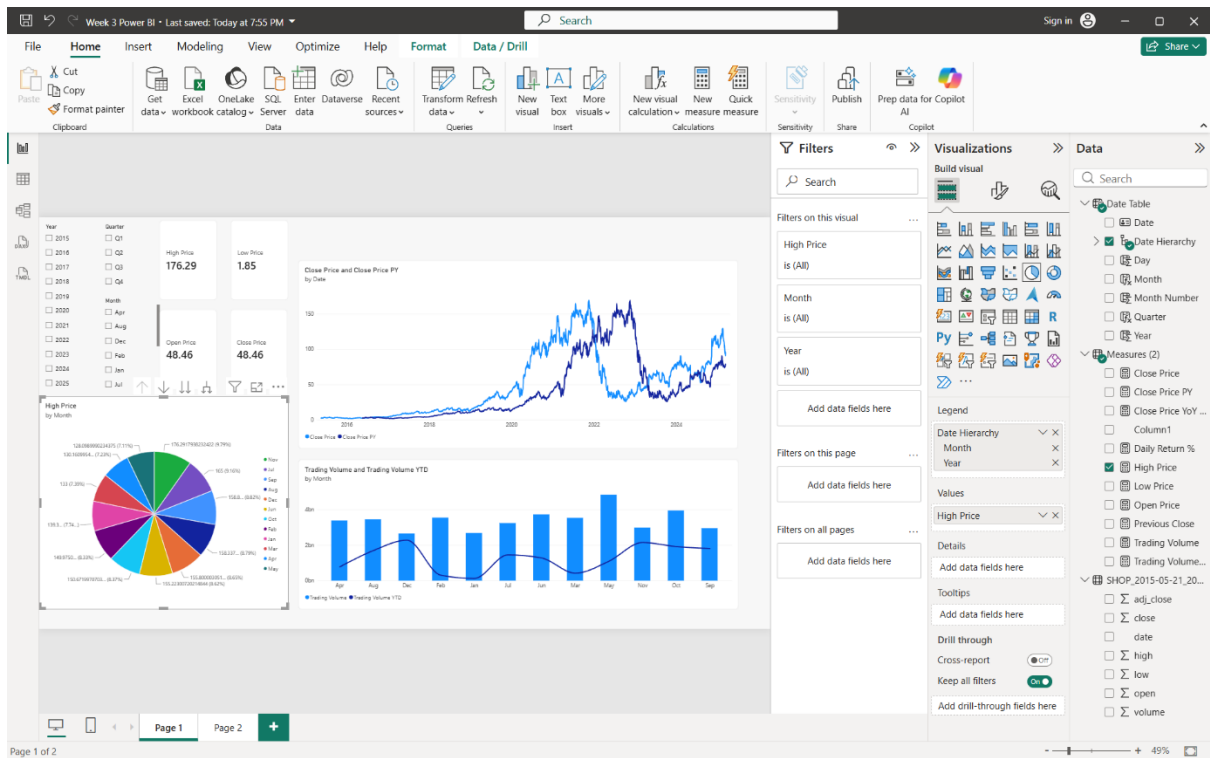
## 6. Visualizations & Insights

### Dashboard was built with 4 sections:

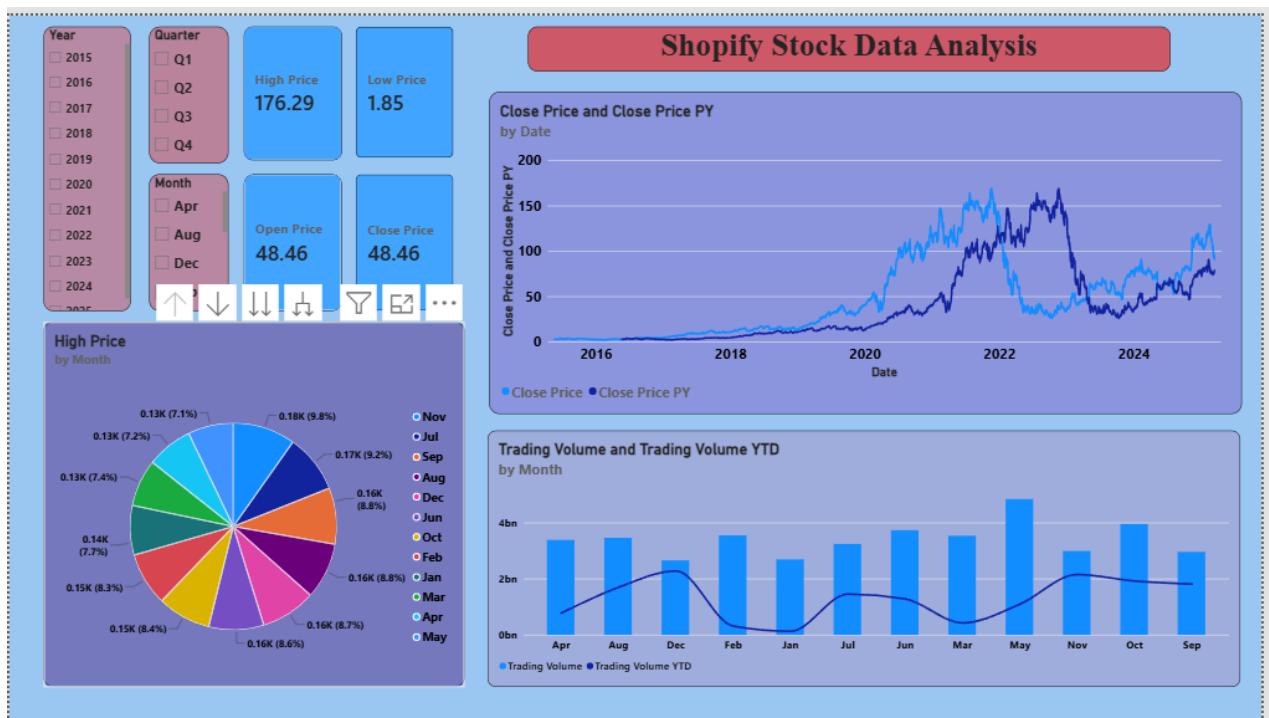
- KPI Cards: Shows latest High , Low , Open and Close price.
- Close Price & Close Price PY (Line Chart) by date: Shows consistent upward trend from 2016 to 2024.
- Trading Volume by Year (Line and stacked column chart)
- High Price by month(Bar Chart)
- Daily Return % (Line Chart) by year: Average daily return showing stable performance.
- Open Price & Close Price by Year(clustered column chart).

**Date Hierarchy (Year-Quarter-Month-Day) was implemented for drill-down analysis.**





## 7. Final Dashboard:





## 8. Conclusion:

The model is now optimized for financial analytics. By using Date Table and time intelligence functions like DATEADD, TOTALYTD and SAMEPERIODLASTYEAR, we can easily compare YTD and YoY performance. The dashboard provides a clear view of Shopify's stock performance over time and trading activity.